



TensorDB

为“AI+大数据”应用创新赋能的向量数据库软件

上海爱可生信息技术股份有限公司



向量数据库

int, float,
string, ...

text

json

image
video
audio

domain specific

0 1 2 3 4
5 6 7 8 9

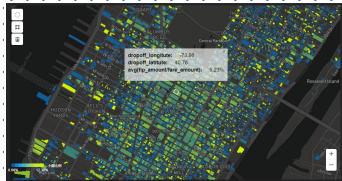
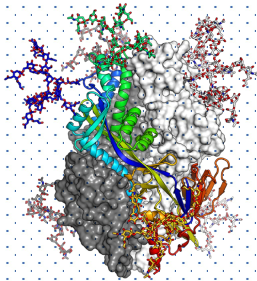
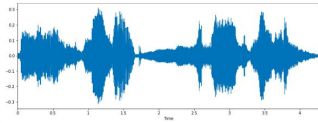
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ABCDEFGH

2021.04.10

Abstract
Bigtable is a distributed storage system for managing structured data that is designed to scale to a very large size: petabytes of data across thousands of commodity servers. Many projects at Google store data in Bigtable, including web indexing, Google Earth, and Google Finance. These applications place very different demands on Bigtable, both in terms of data size (from URLs to web pages to satellite imagery) and latency requirements (from backend bulk processing to real-time data serving). Despite these varied demands, Bigtable has successfully provided a flexible, high-performance solution for all of these Google products. In this paper we describe the simple data model provided by Bigtable, which gives clients dynamic control over data layout and format, and we describe the design and implementation of Bigtable.

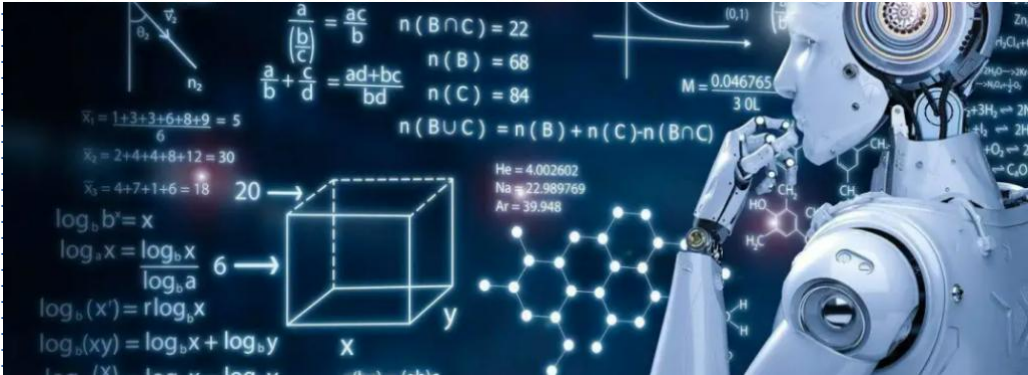
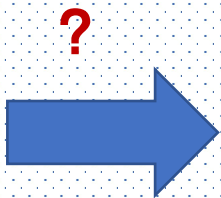
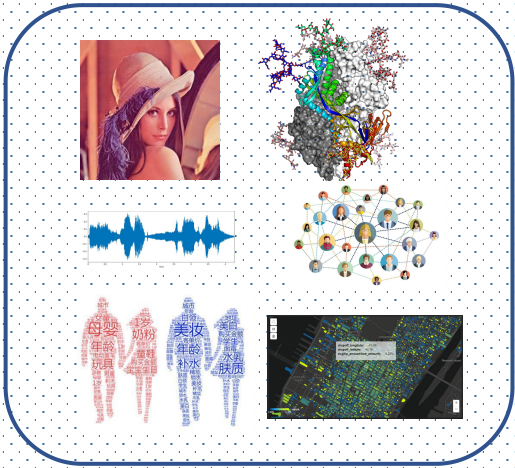
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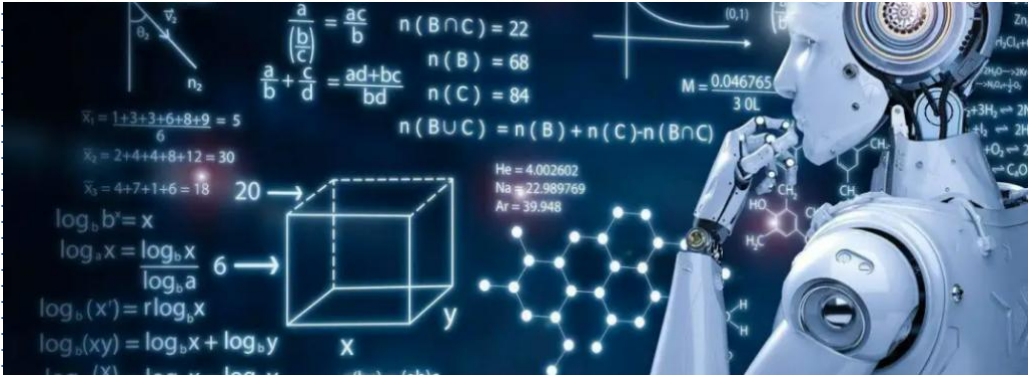
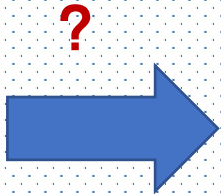
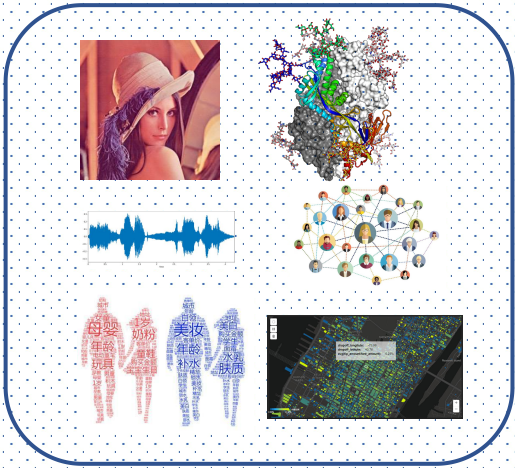
Structured data

Unstructured data

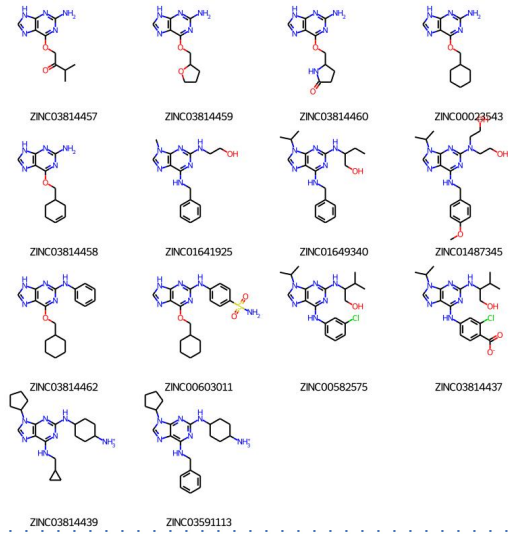
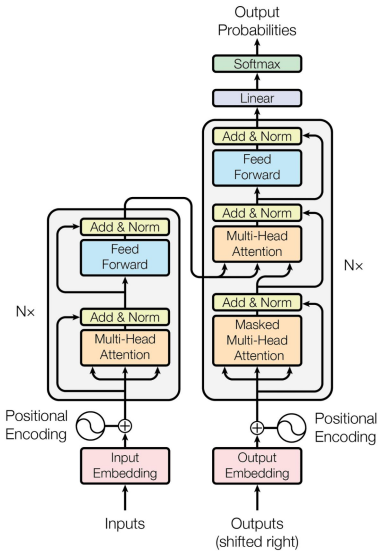
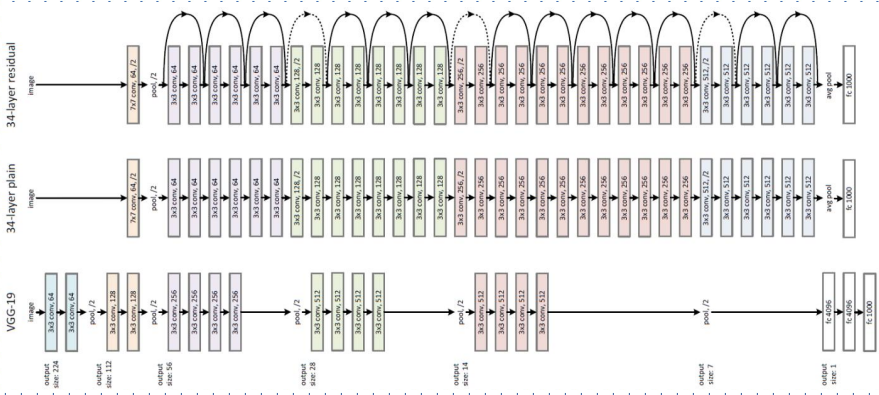
机器理解数据



机器理解数据

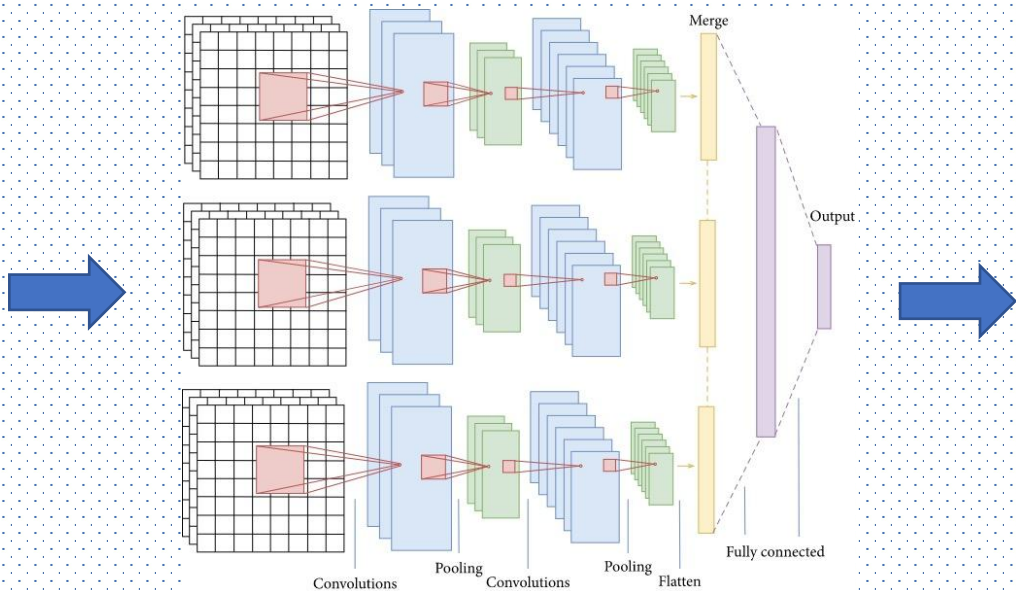


AI模型 / 编码手段

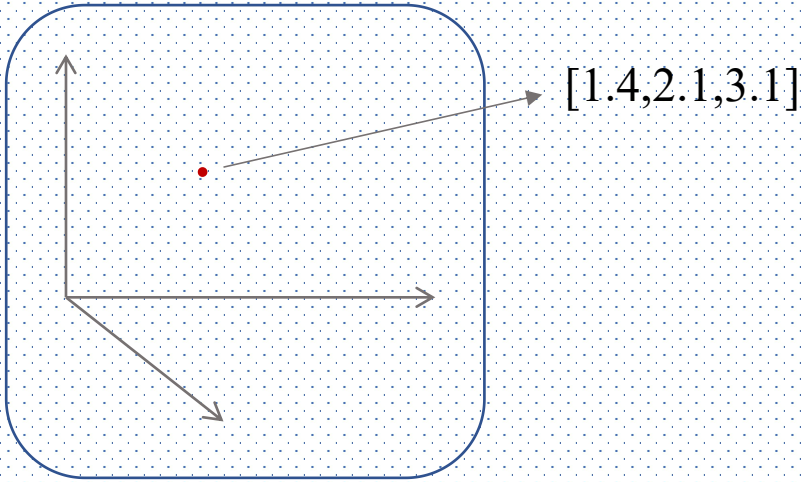


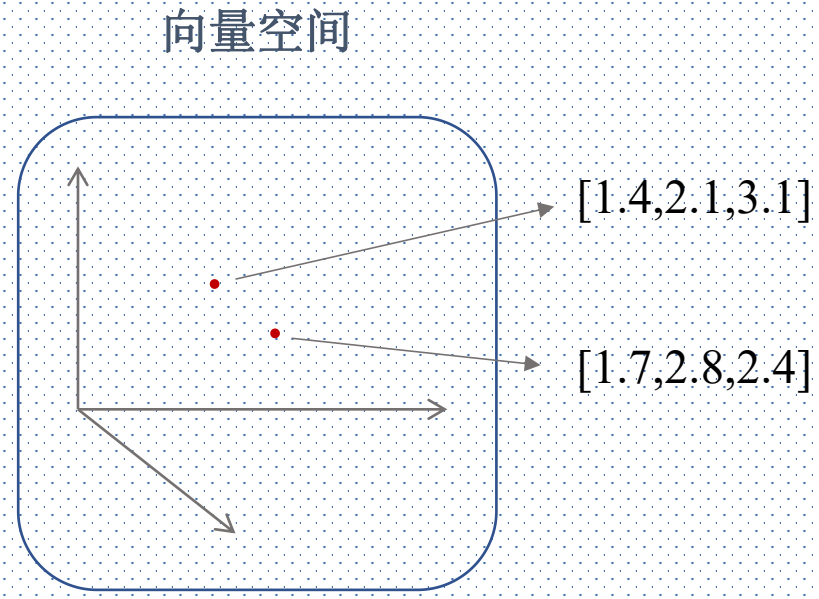
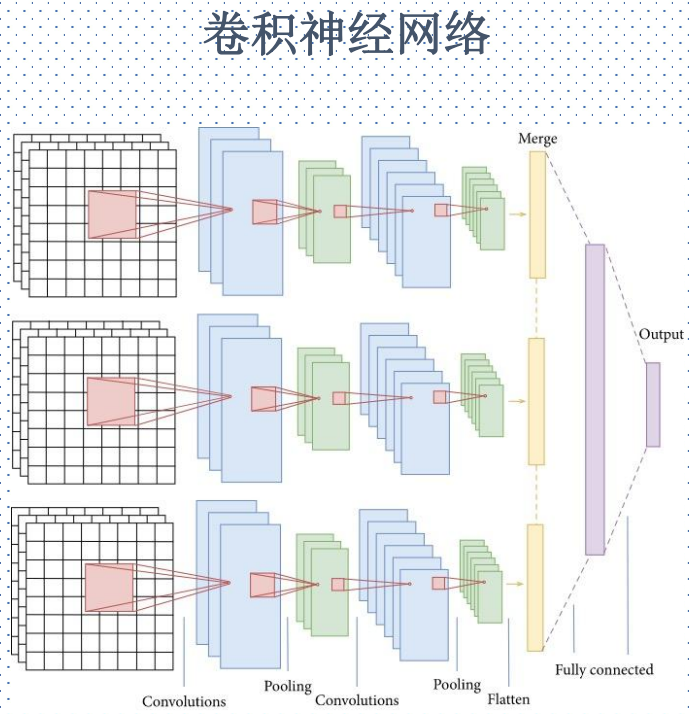


卷积神经网络



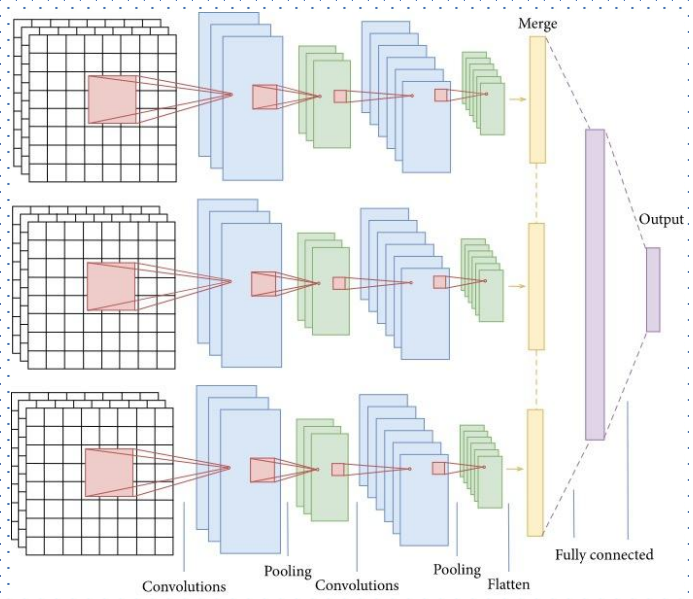
向量空间



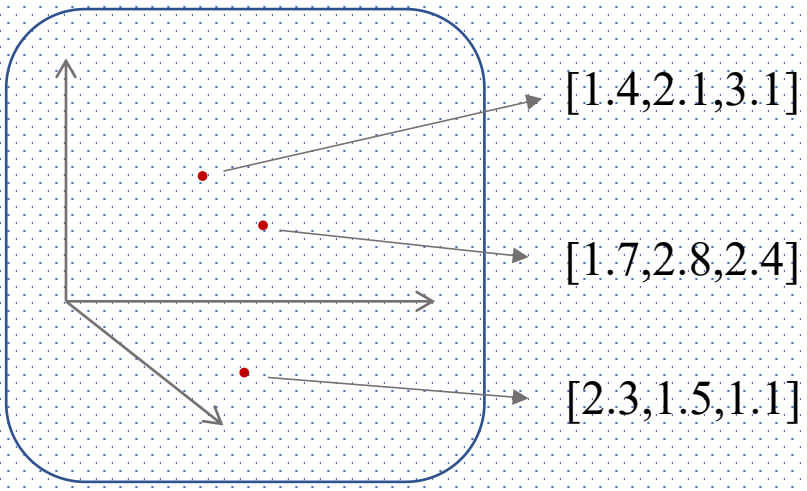




卷积神经网络

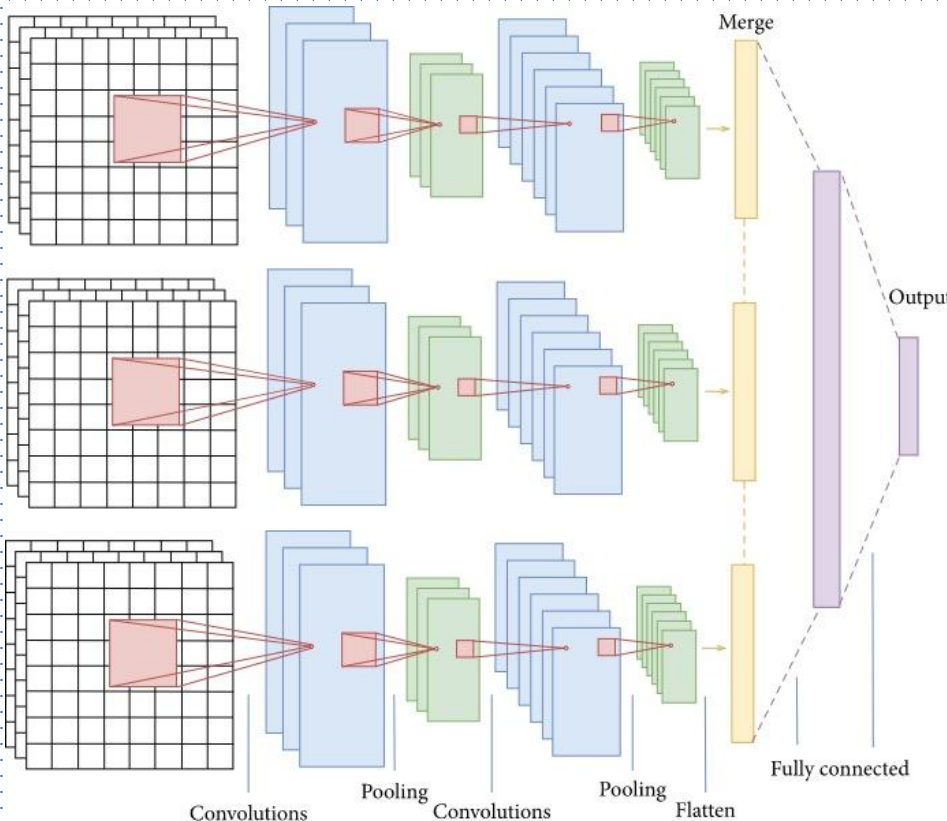


向量空间

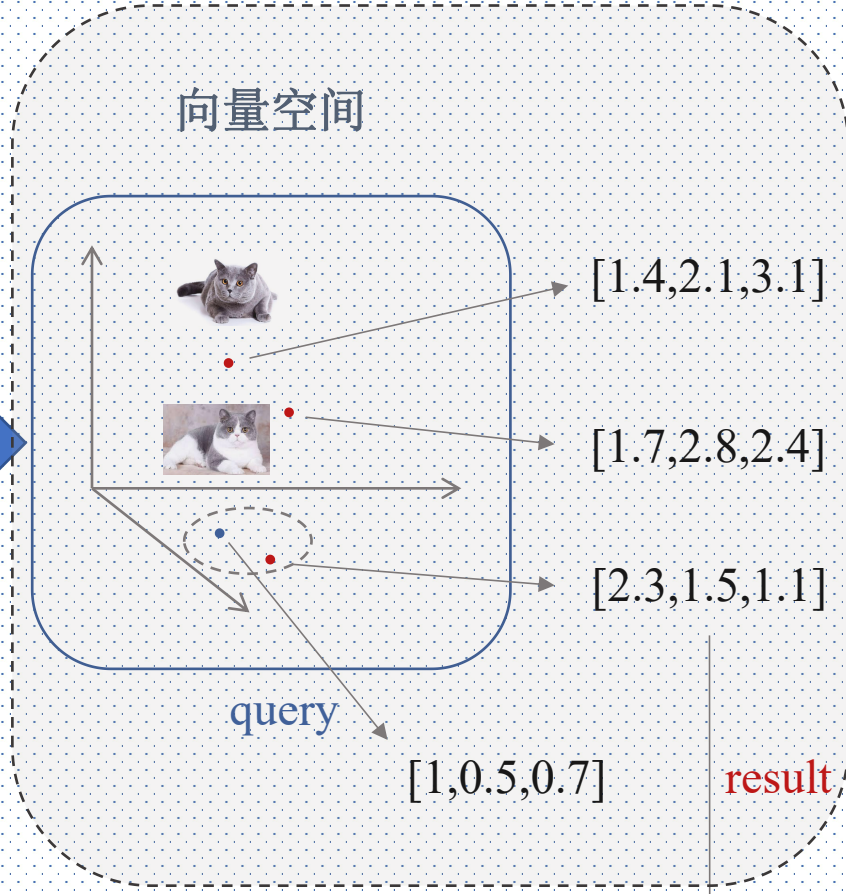




卷积神经网络



向量数据库



query





什么是向量数据库

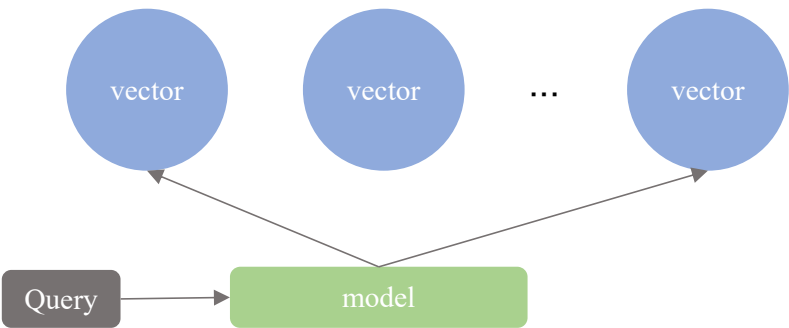
在常见的 AI 场景中，例如以图搜图，视频推荐或者文本检索，需要存储的不是仅仅是图片，视频或者文本片段等非结构化数据，还要存储通过 AI 模型抽象提取出来的高维向量，这些高维向量类似于数学领域中的坐标系的点，标识了不同数据之间的关系，通过向量可以实现数据之间高效的相似性比对，来赋能相应的业务场景。**向量数据库就是可以使用户更加便捷的访问，检索，管理，组织向量的一种数据库。**

TensorDB



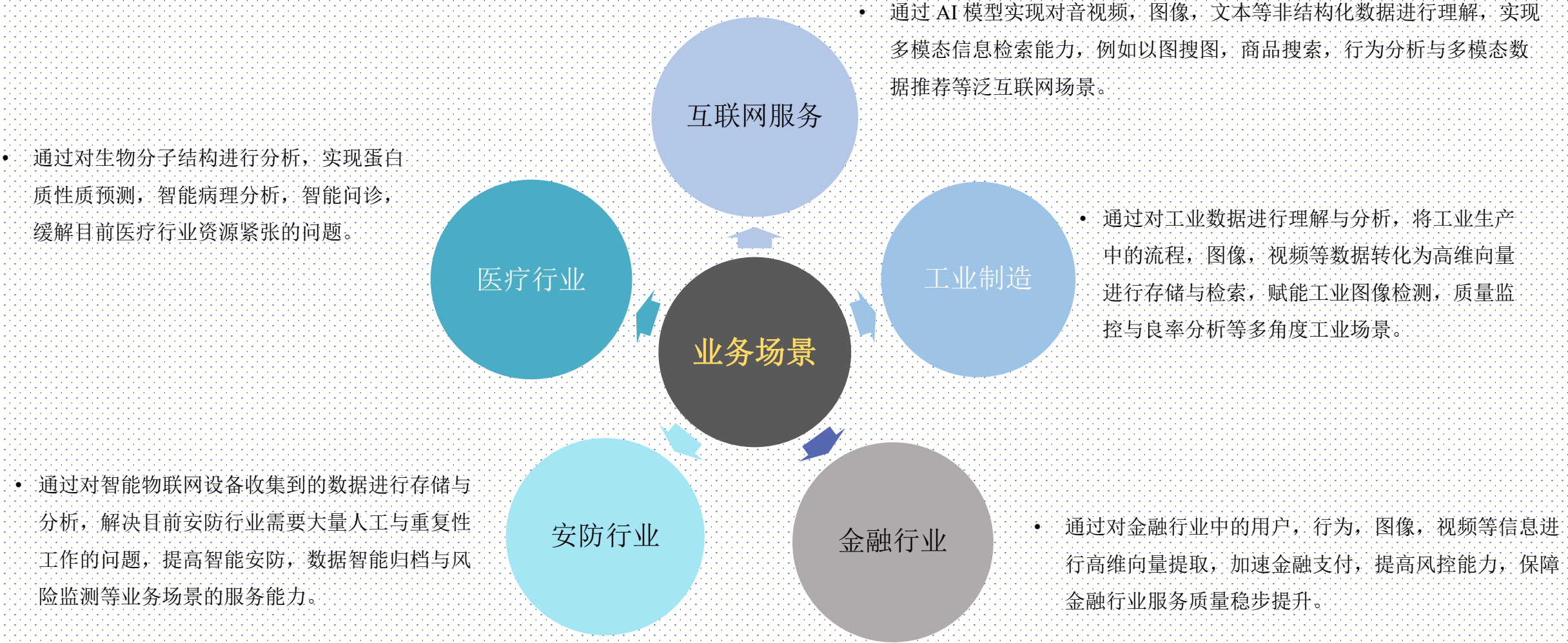
通过大量场景测试爱可生沉淀出针对向量数据库不同索引类型的性能调优方法论，并基于此实现了**自动化调参工具**，帮助用户得到场景上的最优效果。

学习方法





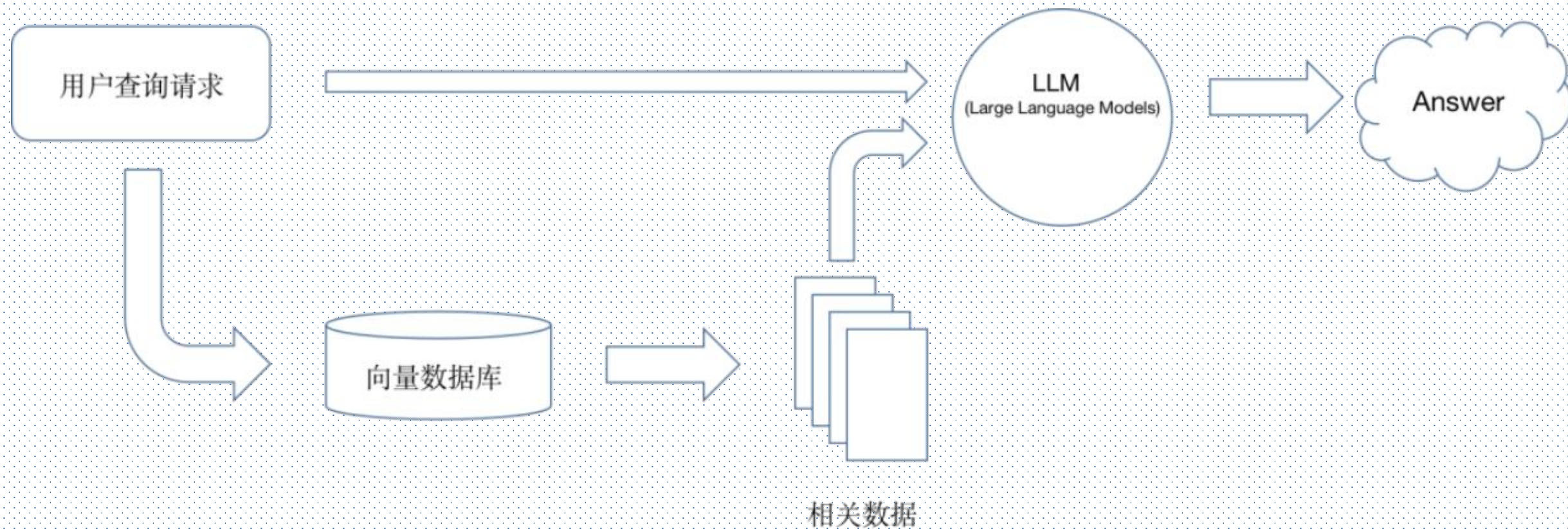
应用场景



舆情看管

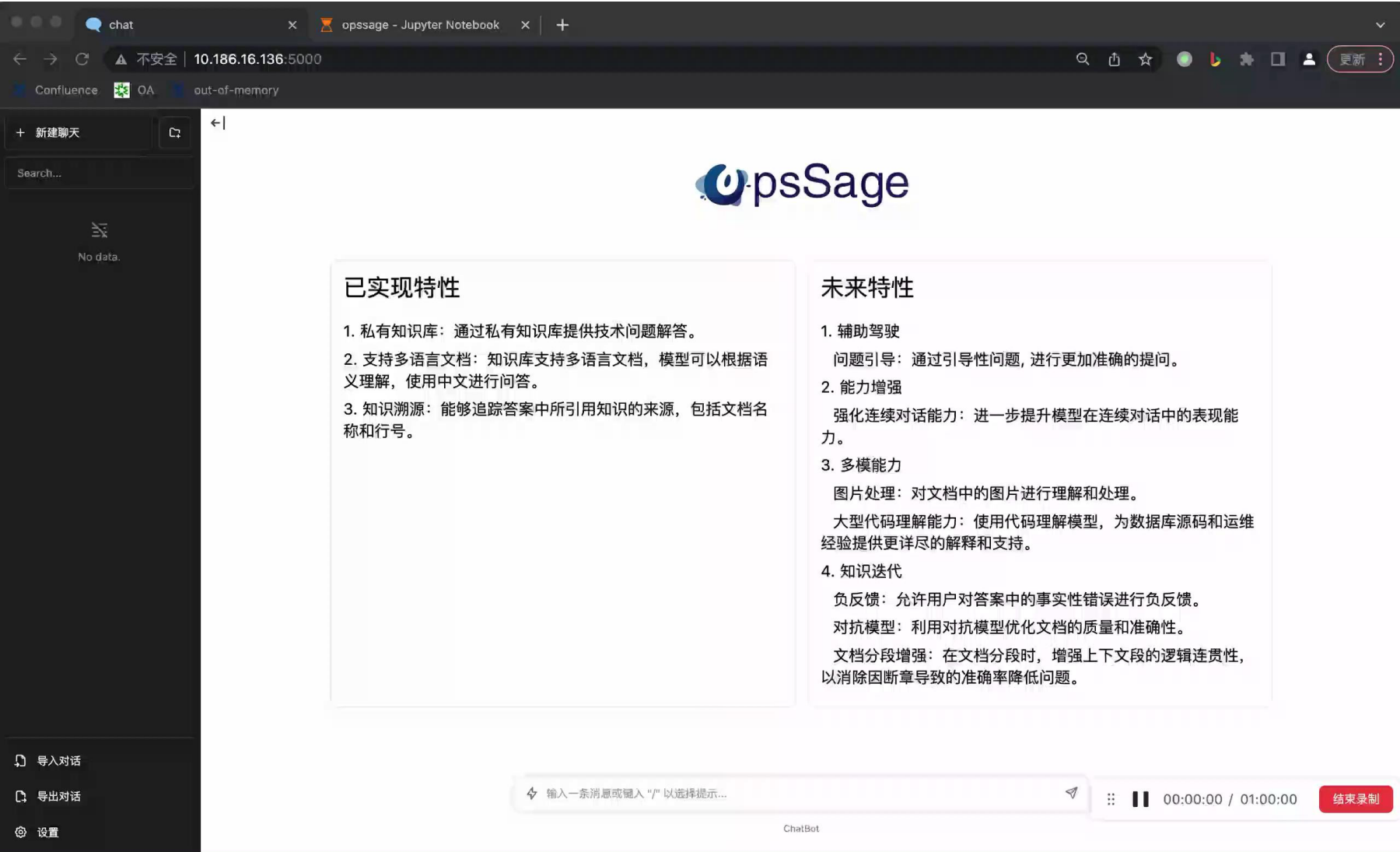


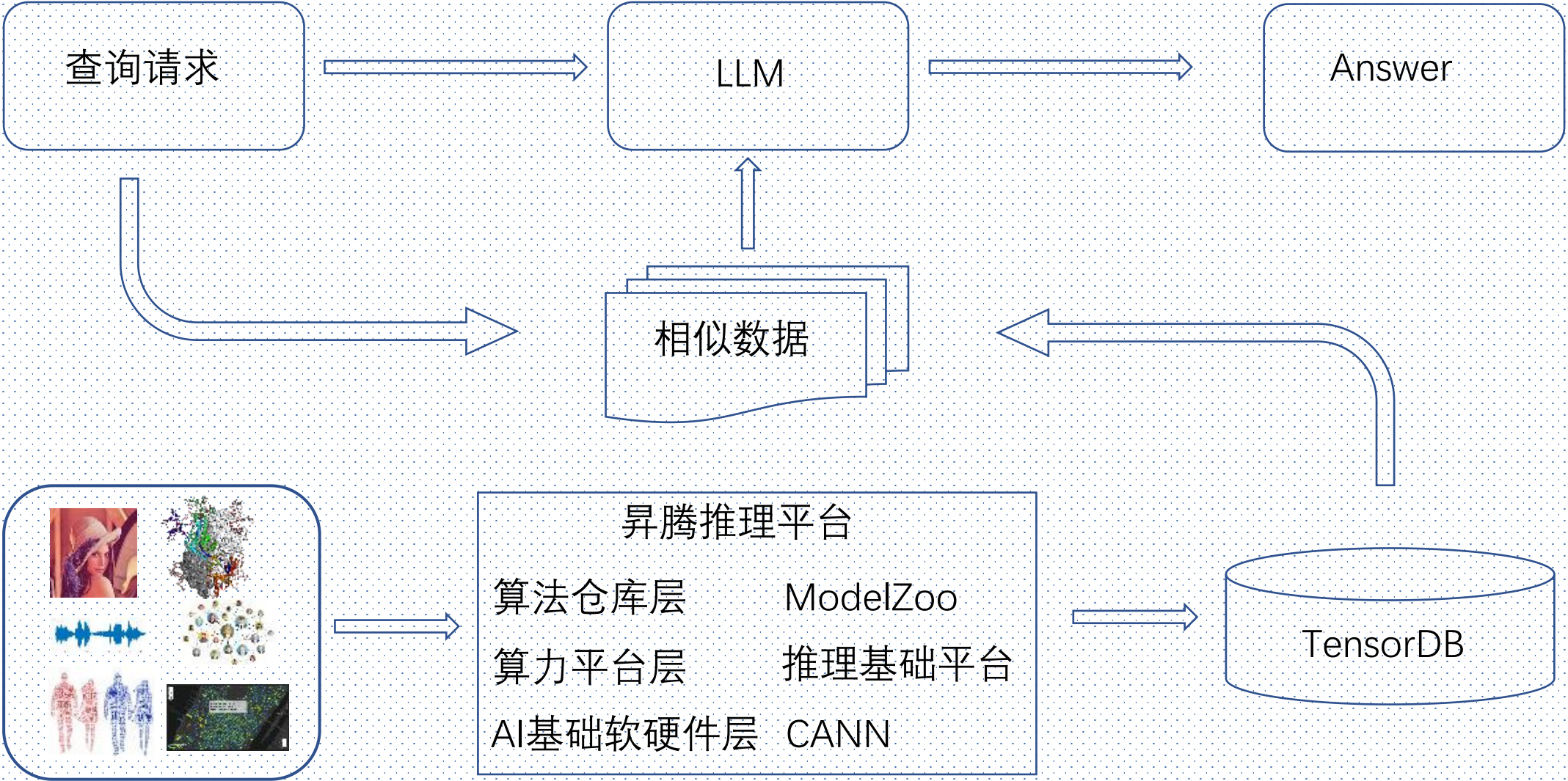
本地知识库



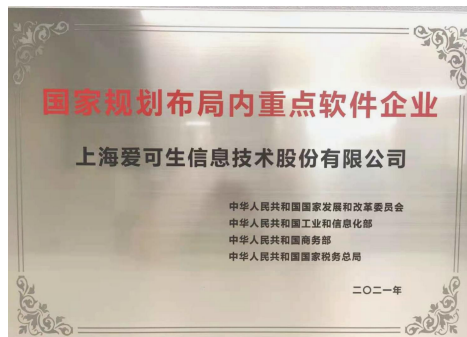


产品案例





公司简介



- 国家规划布局内的重点软件企业、国家高新技术企业。
- 国家科技部国家重点研发计划-“网络协同制造和智能工厂”重点专项。
- 已拥有80多项软件著作权和产品登记证书和**5项已授权核心技术发明专利**（另45+项数据技术专利申请中）。
- 核心产品为多库融合、数据智能应用等产品。
- 数据库标杆客户：工商银行、招商银行、交通银行、农业银行、兴业银行、中国人寿、平安集团、太平洋保险、中国人保、国家电网、上汽集团、上海大众、中国移动、中国电信、华为等40+世界500强，国计民生核心业务的后台数据技术。
- 承担了工信部5所、工业互联网创新中心（上海）、航天科技集团、国家电网等多个重点数据项目。
- 2020年，公司获中国信息协会2020-2021年度信息技术应用创新优秀解决方案奖。
- 2021年上海市重点支持的信创-基础软件企业。



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